

# University Venue Booking System

A campus-scoped venue reservation platform for Mzumbe University — what it does, who uses it, how access and approvals are structured, and how CRs and Admins are kept informed.

**Campuses** Morogoro Main, Dar es Salaam, Tanga, Mbeya · **Platforms** Web application, installable PWA, native mobile app · **Companion documents** User Manual, Admin Manual

## 1 Purpose & scope

The system replaces informal, first-come arrangements for booking lecture halls, laboratories, seminar rooms, and halls with a single reservation record per venue, checked automatically against the official teaching timetable so a booking can never collide with a scheduled class. Each of the university's four campuses runs on the same platform but operates as an isolated unit — venues, timetables, bookings, and people registered on one campus are never visible from another, except to a Super Admin managing the university as a whole.

4

CAMPUSES SERVED

3

ACCOUNT ROLES

2/day

FREE AUTO-APPROVED BOOKINGS

2

NOTIFICATION AUDIENCES

## 2 Who uses it

| ROLE           | WHO                                 | ACCESS                                                               |
|----------------|-------------------------------------|----------------------------------------------------------------------|
| Class Rep (CR) | A registered student representative | Own bookings; own campus's venues, timetable, and notifications      |
| Admin          | Staff managing one campus           | Everything for that campus only, including its Admins' announcements |
| Super Admin    | Staff managing the whole university | All four campuses; also manages other Admins                         |

A CR self-registers; their login email is generated automatically from their name and registration number rather than chosen by them, tying every account to a verifiable student identity. Admin and Super Admin accounts are created by an existing Super Admin — there is no public sign-up path into either.

### 3 The booking workflow

A CR chooses a semester, date, and time window; the system returns only venues that are genuinely free — cross-checked against both the official timetable and every other CR's existing booking for that slot — so what's offered can never actually clash. A digital signature, drawn on screen, is required to confirm.

To keep venues available fairly across a busy campus, each CR is auto-approved for up to **two bookings per calendar day**. A third request on the same day requires a written reason and is held as **Pending** for their campus Admin to approve or reject — the same review queue Admins already use for anything needing a human decision. Once decided, the CR is notified immediately, and if it was reviewed by a person rather than the system, the record shows exactly who approved it.

#### WHY A LIMIT AT ALL

Unrestricted same-day booking by a small number of CRs was capable of exhausting a campus's available venues; the two-a-day allowance covers ordinary use without any review step, while a documented reason is asked for anything beyond it.

### 4 Feature overview, by area

#### Venues

Add, edit, and retire venues per campus, with capacity, type, and optional restrictions by booking purpose, academic level, or department. Long lists (CR and Admin views alike) reveal progressively via a "Show More" control rather than rendering everything at once.

#### Timetable import

Official lecture schedules are pulled directly from the university's published timetable (by link) or a CSV file, per semester and campus, in either additive or replace mode.

#### Bookings

A searchable, filterable review queue for every booking — approve, reject with a reason, or delete — plus a PDF report scoped to a status and, optionally, a single date.

#### Booked Venues

A day-by-day or full-history read view of what's occupying every venue, including each booking's captured signature. The CR-facing day view also grows progressively via "Show More"; the Admin history view instead uses page-number pagination, matching every other Admin list.

#### Semesters

Defines the academic terms bookings and timetable imports are organized under; exactly one is active at a time.

#### Students (CR) management

Individual or bulk (CSV) registration, editing (including correcting a Reg No after the fact), and removal. Removal is a soft delete — personal details are cleared, but booking history is preserved under a placeholder identity.

#### Admin management

Super Admins create and manage Admin accounts, assign each to a campus, and may promote another Admin to Super Admin. One original account — the Main Super Admin — is permanently protected from edits, demotion, or removal by anyone else.

## Notifications

A persisted, per-recipient notification record with read/unread state — not a client-side approximation. A CR is notified the moment a booking is approved or rejected; an Admin is notified the moment a CR's booking needs review, with a direct link into the review queue. An unread count badge is shown on a bell icon in the header at all times, updating live.

## Announcements

Any Admin can broadcast a message either to CRs (their own campus, or a chosen/every campus for a Super Admin, optionally narrowed by faculty, department, program, level, or year of study) or directly to fellow Admins. Every Admin can see what's been announced on their own campus — by any Admin, not only themselves — along with how many people received it and how many have read it, though editing or deleting a given announcement stays restricted to whoever originally posted it.

## Activity logs

A record of logins, bookings, and administrative actions, filterable by action type. A Super Admin's own actions are visible only to that Super Admin, keeping day-to-day Admin oversight uncluttered.

## Settings & branding

System color and logo (any Admin); app name, support phone, footer, and login-page appearance (Super Admin only); per-weekday booking hours for Study Unit purpose bookings.

## Offline support

Installable as a home-screen app (and as a native mobile app); previously viewed pages and data remain readable with no connection, while actions that write to the server are clearly flagged as needing one.

## 5 Booking status glossary

| STATUS    | MEANING                                                                                              |
|-----------|------------------------------------------------------------------------------------------------------|
| Pending   | Awaiting a campus Admin's decision — always a 3rd-or-later same-day booking, with a reason attached. |
| Approved  | Confirmed automatically by the system, or by an Admin after review.                                  |
| Rejected  | Declined by an Admin, with a reason recorded for the CR.                                             |
| Cancelled | Withdrawn by the CR themselves while still Pending.                                                  |

## 6 Roles & permissions summary

| CAPABILITY                       | CR            | ADMIN      | SUPER ADMIN |
|----------------------------------|---------------|------------|-------------|
| Book a venue on their own campus | Yes           | —          | —           |
| Approve / reject bookings        | —             | Own campus | Any campus  |
| Manage venues & timetable        | —             | Own campus | Any campus  |
| Manage CRs                       | —             | Own campus | Any campus  |
| Manage Admin accounts            | —             | —          | Yes*        |
| Post / view campus announcements | Receives only | Own campus | Any campus  |
| Change system-wide branding      | —             | —          | Yes         |
| Apply database updates           | —             | —          | Yes         |

\*Except the Main Super Admin, protected from edits, demotion, and removal by anyone but themselves.

## 7 Availability & access

The application is reachable as a standard web page and can additionally be installed to a phone's home screen as a Progressive Web App, or used through a dedicated native mobile app. Once installed and opened online at least once, previously viewed pages and data remain available with no connection; the interface indicates offline status plainly rather than allowing an action to fail silently. All interface text is presented in English across both the Class Representative and Administrator experiences.

## 8 Data model (ER diagram)

The core entities and how they relate. Every foreign key that can be left empty (e.g. a Notification that isn't tied to a specific booking) is drawn as a dashed line; every table also carries standard `created_at`/`updated_at` timestamps, omitted below for clarity.

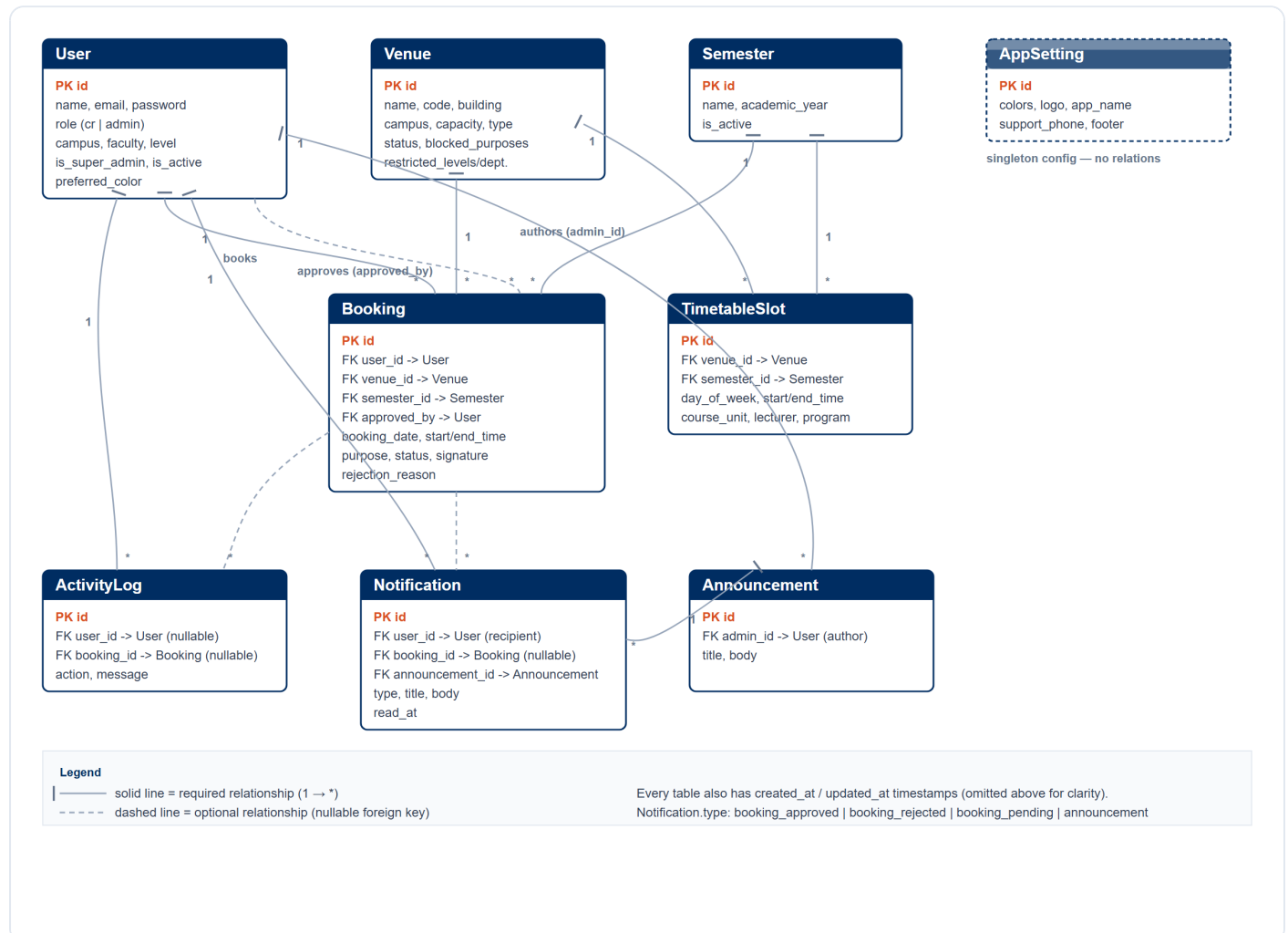


Figure 1 — Entity Relationship Diagram

## 9 Use case diagram

What each account role can actually do. A Super Admin's capabilities are a strict superset of a regular Admin's (shown as the "extends" relationship) — anything an Admin can do, a Super Admin can also do, across every campus rather than just their own.

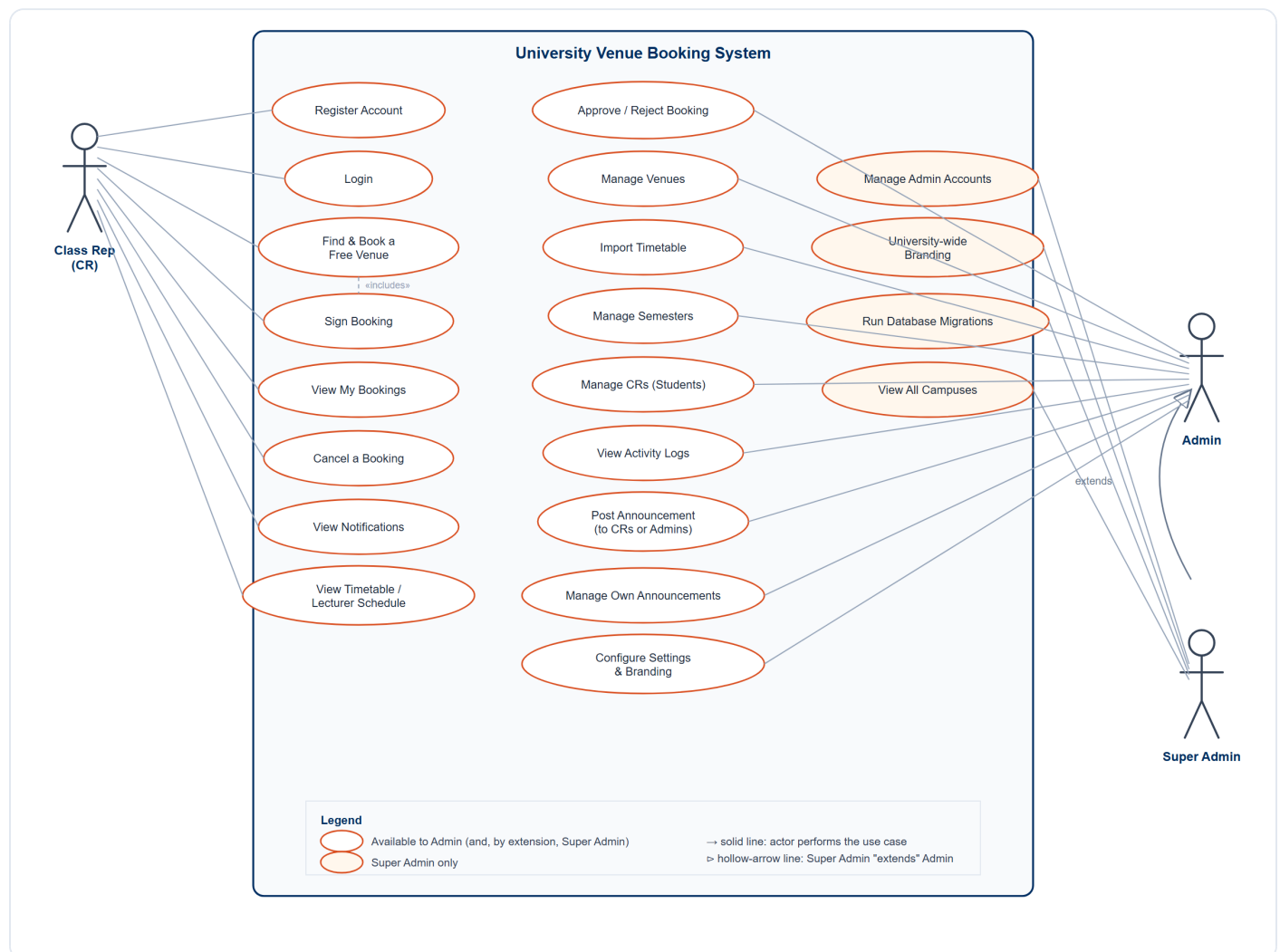


Figure 2 — Use Case Diagram

## 10 Related documents

- **User Manual** — step-by-step instructions for Class Representatives: registering, booking a venue, tracking bookings, using notifications, and installing the app.
- **Admin Manual** — step-by-step instructions for Administrators and Super Admins: venues, timetable, bookings, CRs, other Admins, notifications, announcements, and system settings.